EF119 – Macroeconomics
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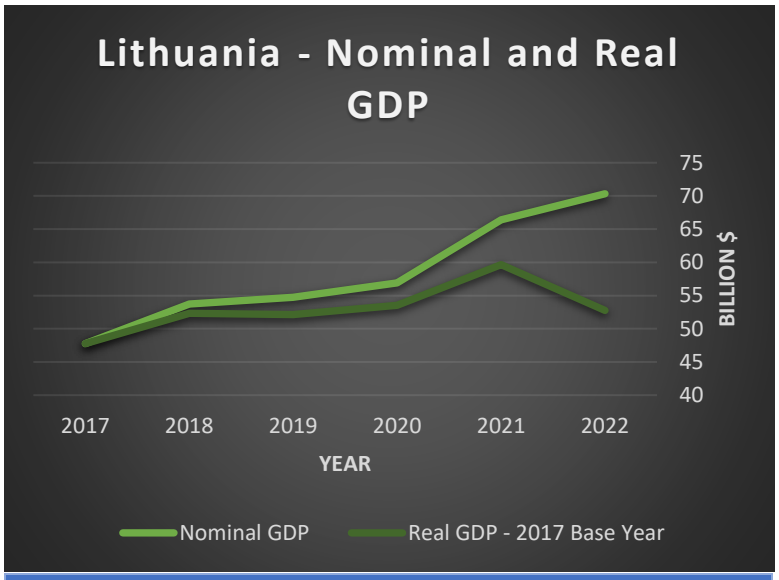
Question 1

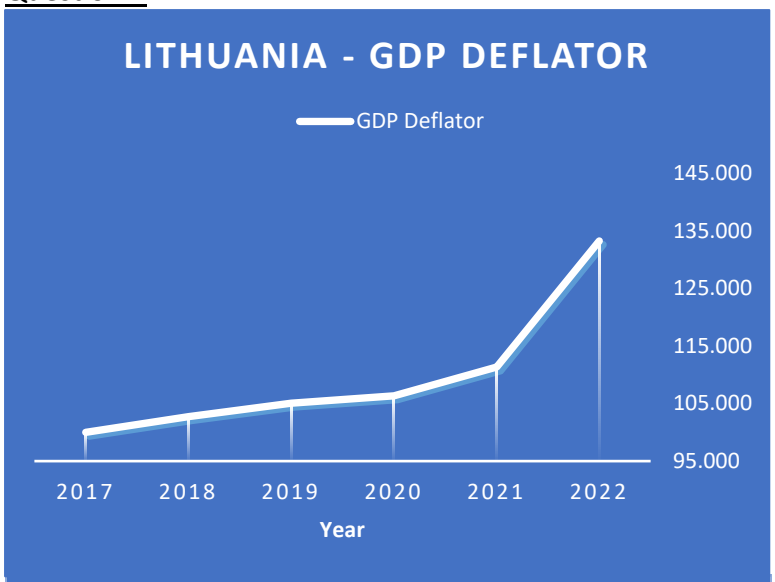


Year	Nominal GDP (Billion $)	Inflation Rate + 1	Adjustment for 2017 base year (Inflation 2019 x 2018 x 2017, etc)	Nominal GDP/(1+Inflation Rate) = Real GDP (Billion $) with 2017 Base Year
2022	70.33	1.197	1.333	52.780
2021	66.41	1.047	1.113	59.657
2020	56.91	1.012	1.063	53.526
2019	54.76	1.023	1.051	52.122
2018	53.75	1.027	1.027	52.337
2017	47.76	1	1.000	47.760

I have chosen Lithuania as the country to do my research of GDP on. I gathered the values of Nominal GDP and Inflation Rates from https://data.worldbank.org/ and proceeded to find the Real GDP values with base year 2017.

From my research I discovered that Lithuania's nominal GDP rises steadily. This indicates constant economic growth, due to an increase of goods and services being produced and sold within Lithuania over this 6 year period. On the other hand, real GDP also grows at a steady rate up until 2022. Up until this drastic decrease the values indicate positive economic growth, similarly to nominal GDP. The large decrease indicates that Lithuania may not have been able to sustain their constant economic growth. This can be due to a number of factors such as unemployment, a decrease in economic activity or high inflation prices. According to Motley (1998), times of high inflation rates lead to decreased real GDP. It is clear that this is the case for Lithuania.

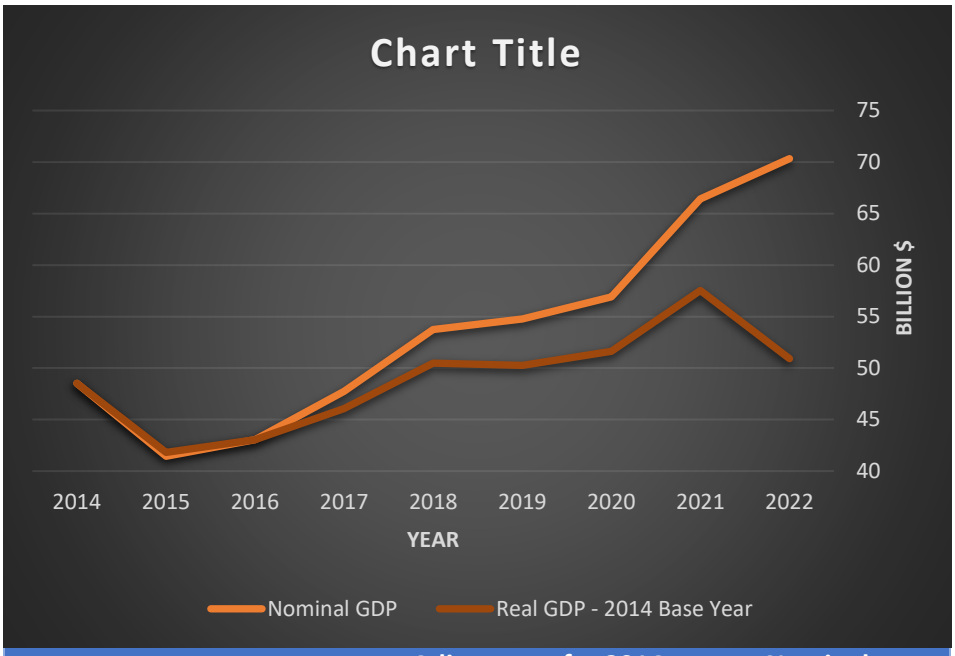
Question 2



Year	Nominal GDP (Billions $)	Real GDP (Billion $)	Nominal/Real x 100 = GDP Deflator
2022	70.33	52.780	133.250
2021	66.41	59.657	111.320
2020	56.91	53.526	106.323
2019	54.76	52.122	105.062
2018	53.75	52.337	102.700
2017	47.76	47.760	100.000

I calculated the GDP deflator using the values of nominal and real GDP.
From the graph on the left, it is obvious that the GDP deflator continuously increases year on year. This steady increase indicates that all aspects of Lithuania's economy has been growing steadily. This may include areas such as price of goods/services, wages, inflation rates, interest rates and exchange rates.

Question 3



Year	Nominal GDP (Billions $)	Inflation Rate + 1	Adjustment for 2014 base year (Inflation 2018 x 2017 x 2017 x 2016 x 2015 x 2014, etc)	Nominal GDP/(1+Inflation Rate) = Real GDP (Billion $) with 2014 Base Year
2022	70.33	1.197	1.382	50.901
2021	66.41	1.047	1.154	57.533
2020	56.91	1.012	1.102	51.620
2019	54.76	1.023	1.089	50.266
2018	53.75	1.027	1.065	50.474
2017	47.76	1.037	1.037	46.060
2016	43.05	1.009	1.000	43.053
2015	41.44	0.991	0.991	41.816
2014	48.53	1	1.000	48.530

Similarly to Question 1, I have used the values of nominal GDP and inflation rates to find the real GDP with a base year of 2014.

When 2014 is used as the base year, we can identify that in 2015 and 2016 the real GDP exceeds the nominal GDP. This can only occur due to deflation in the economy. In 2015, Lithuania adopted the Euro as its currency which may have been the root cause of why deflation that was suffered. Deflation has been a trend for the vast majority of eurozone members when they first joined (Grabbe, 2014). This is due to exchange rates and price adaptations.

References

Grabbe, H. (2014). Six Lessons of Enlargement Ten Years on: The EU's Transformative Power in Retrospect and Prospect. *Journal of Common Market Studies*, *52*, 40–56. https://heinonline.org/HOL/P?h=hein.journals/jcmks52&i=1451

Motley, B. (1998). *Growth and Inflation: A Cross-Country Study*. *1*.

World Bank Open Data. (n.d.). World Bank Open Data. Retrieved October 19, 2023, from https://data.worldbank.org